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Oligo-Miocene rotation of Sardinia: K-Ar ages and paleomagnetic data of Tertiary volcanics

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Conventional K-Ar dating was carried out on mineral separates, mostly plagioclases, from Middle Tertiary volcanics of Sardinia along with paleomagnetic studies. the close agreement of dates from the two methods establishes the rotation age of Sardinia relative to Europe within a rather short time span of 1.5 Ma between 20.5 and 19 Ma. From correlation of our results with the geomagnetic time scale, it is suggested that the rotation took place shortly after the magnetic anomaly 6.

1. Introduction

An intense rifting process, that gave rise to a series of grabens cutting the Hercynian basement and a widespread calc-alkalic volcanism, featured the geological history of Sardinia in the Tertiary. The volcanic activity stopped during the Langhian and resumed later in the Plio-Pleistocene. This second volcanic episode was of alkalic nature and rather short-lived compared to the earlier calc-alkalic one.

Numerous paleomagnetic studies were carried out on Tertiary volcanics in the last twelve years, to scrutinize a possible rotation of Sardinia relative to Europe in Oligo-Miocene times. Some concluded to a counterclockwise rotation in Oligo-Miocene [1–4], while others challenged this hypothesis [5,6]. More recent investigations based on data from about 100 sites, however, favored the counterclockwise rotation [7–9] and ascertained the event at 16–17 Ma age. Reservation had to be made on this conclusion since the samplings for K-Ar dating and paleomagnetic investigations had been performed independently. Serious doubts arose whether both informations pertained to the same object or not. Furthermore, the ignimbritic series was almost ignored in those correlations.

The aim of the present work is two-fold: (1) to discuss the numerous available paleomagnetic data from a statistical point of view in order to assess the amplitude of the rotation; (2) to determine the time of the rotation by K-Ar dating on samples from the same sites where paleomagnetic studies were carried out.

2. Paleomagnetic data

Most of the paleomagnetic data previously published [9] are presented in Table 1. Measurements from five sites (sites 143, 145, 146, 147, 148), though, represent new results. The outlines of sampling and analytical techniques are, therefore, briefly recalled.

2.1. Techniques

The sampling attempted to cover the largest stretch of geological time and the widest geographical area. 98 sites of cal-alkalic andesites, ignimbrites and tuffites were sampled in Anglona, Gallura, Logudoro, Bosano, Cixerri and Sulcis (Fig. 1). Aquitanian and Langhian tuffites, as M_1t

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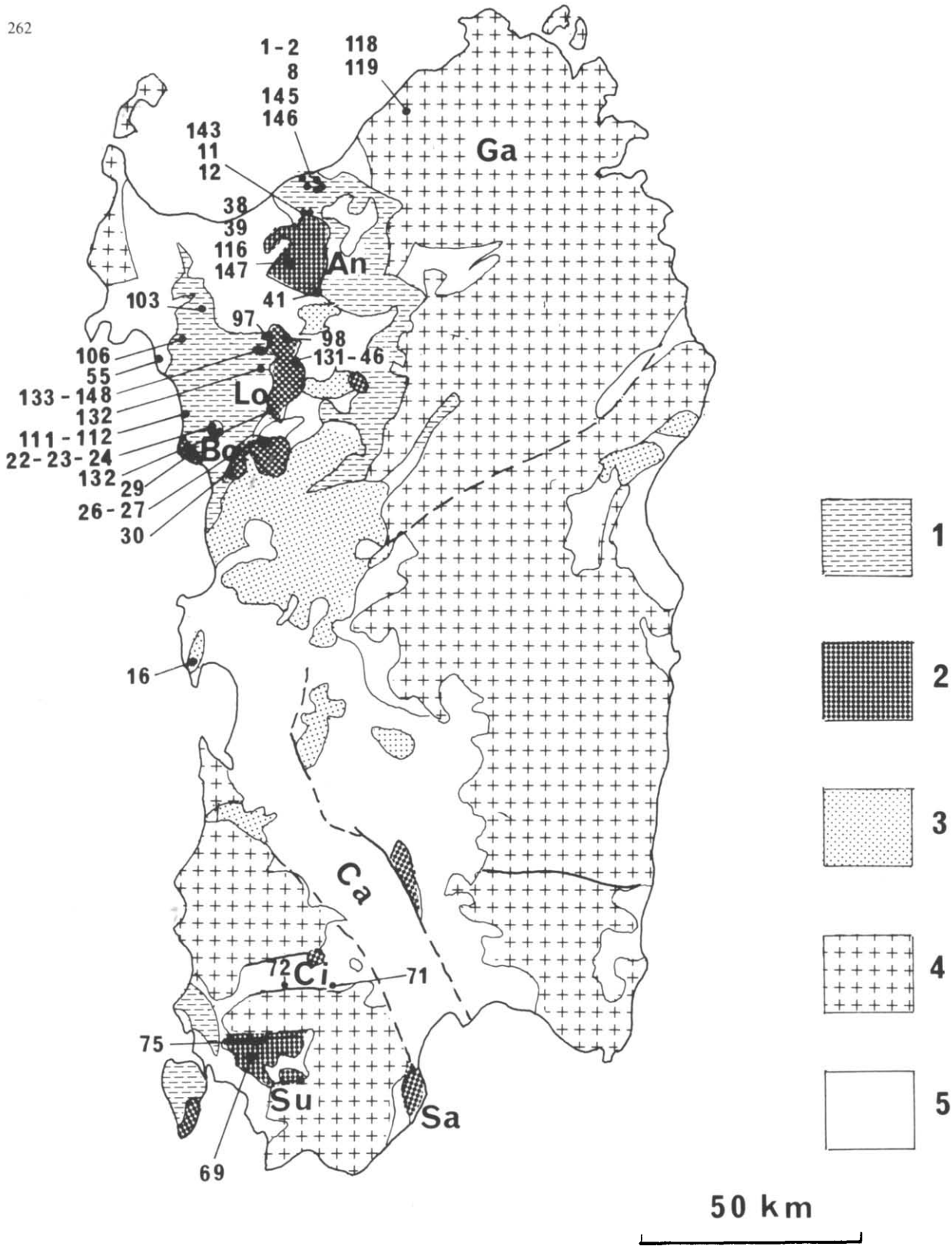


Fig. 1. Volcanics of Sardinia and location of sampling sites [9,10] taken into account for paleomagnetism and K-Ar dating. *1* = Oligo-Miocene ignimbrites, *2* = Oligo-Miocene andesites, *3* = Plio-Pleistocene basalts, *4* = Paleozoic basement, *5* = sediments. An = Anglona, Ga = Gallura, Lo = Logudoro, Bo = Bosano, Ca = Campidano, Ci = Cixerri, Su = Sulcis.

on the geological map of Castelsardo [14], andesites as α and ignimbrites as τ_2, τ_3 and $M_1\tau$, which is interbedded with M_1t, were taken from Anglona for analysis. All of the formations defined by Coulon et al. [6,26] in Logudoro and Bosano were studied: the lower andesitic series as SA_1, which approximately corresponds to α_1 on the geological maps of Bonorva, the upper andesitic series as SA_2 or α_2, the upper ignimbritic series as SI_2 or τ_2 and the uppermost andesites as SA_3. In Cixerri and Sulcis, the sampling was focused on andesitic domes and dikes, described as α on geological maps.

Thermal treatments indicated that the tuffites and most of the andesites had undergone hydrothermal alteration [9,10]. In most cases, refined alternating current and thermal demagnetizations erased secondary magnetizations. These procedures usually yielded fairly good characteristic magnetizations.

2.2. Results

Our paleomagnetic results are summarized in Fig. 2. Three sites out of 98 showing diverging directions have not been taken into account in the discussion of geomagnetic direction. The remaining sites display directions of characteristic magnetization that are moderately scattered, lying in the NW and SE quadrants. The reversed magnetizations cluster around 140–150°N and the normal ones around 330–340°N (Fig. 2); the mean direction calculated with the high-quality data (N samples >3, $\alpha_{95}<20°$), i.e. 84 sites, is $D=334°$, $I=51°$, $\alpha_{95}=4°$. The earlier published data on the calc-alkalic volcanics confirm the northwest trend of the declinations [1–4,6,11–13]. A statistical approach of the rotation can be attempted by taking into consideration of our results with previously published data [1–13]. Such a treatment is necessary because of the variability in directions. Although the paleomagnetic data are of high quality, with no secondary magnetization and suitable tectonic corrections, the lower ignimbritic series of Logudoro-Bosano (SI_1 or τ_1) shows variable directions [9,10]. This observation suggests that important secular variations and anomalous field directions have occurred during the cooling of the lavas. The numerous field reversals, in the Tertiary, might have thus induced most of those anomalous directions. Only a significant number of sites, therefore, sampled from various levels of the volcanic series, can integrate the field variations.

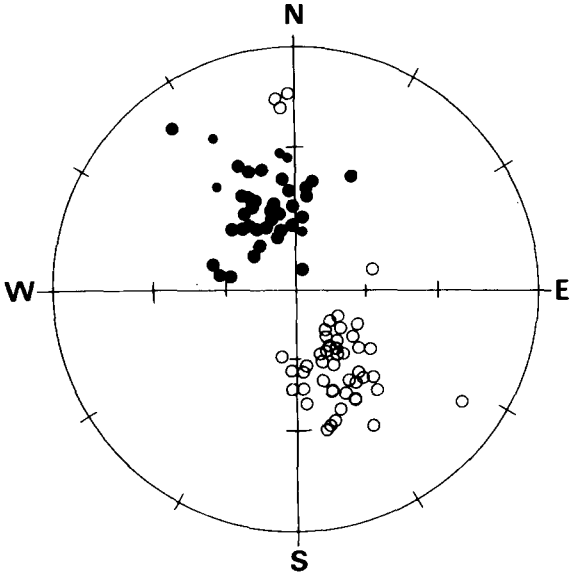


Fig. 2. Characteristic directions of magnetization of the calc-alkaline volcanics [9,10]. Projection of site mean directions: ● = normal polarity, ○ = reverse polarity.

Nearly 200 sites of calc-alkalic volcanics that fulfil the requirements of N samples >3 and $\alpha_{95}<20°$ are taken into account in the statistical approach. The inclinations lie inside a 30–70° range (Fig. 3a), with a maximum of population around 52°. When the directions with low inclinations are discarded, the declination values are distributed into three groups, namely 280–310°, 310–350°, 350–10° (Fig. 3b). The first group mostly represents the same ignimbritic flow $M_1\tau$ sampled by different workers at distinct localities near Castelsardo [2,7,13] and several andesites from southwest Sardinia [9,10]. Similar directions have also been found recently on volcanics from Sarroch [30] (Fig. 1). The second group, represented by a great number of sites ($N=102$), shows a symmetrical repartition. The mean direction for the intervals $30°<I<70°$ and $310°<D<350°$ is $I=51°$, $D=329°$, $\alpha_{95}=2°$. The corresponding

TABLE 1

K-Ar dating and paleomagnetic directions of Sardinian volcanics

N	Locality	Rock type	Min.-eral	K_2O (%)	$^{40}Ar_{rad}$ (10^{-11} mol/g)	$100\,^{40}Ar_{rad}/^{40}Ar_{total}$	Age (Ma) ± 1σ (a)	Age (Ma) ± 1σ (b)	*D* (°)	*I* (°)	α_{95} (°)
3	Castelsardo	$M_1\tau$	pl	0.690	7.255	81.0	69.8 ±2.3	71.6 ±2.3	328	58	3
145	Castelsardo	τ_2	pl	1.222	3.484	62.6	19.2 ±0.5	19.7 ±0.5	354	42	2
			pl	1.210	3.368	67.5	18.7 ±0.5	19.2 ±0.5			
			pl	1.210	3.317	70.5	18.5 ±0.5	18.9 ±0.5			
146	Castelsardo	τ_2	pl	1.230	3.463	70.6	19.0 ±0.5	19.5 ±0.5	7	58	4
			pl	1.210	3.362	69.0	18.7 ±0.5	19.2 ±0.5			
143	Castelsardo	τ_2	pl	1.250	3.488	40.1	18.8 ±0.5	19.3 ±0.5	357	47	3
111	Pedra Ettori	SI_1	–	–	–	–	–	–	141	−61	3
112	Pedra Ettori	SI_1	pl	0.423	1.420	47.0	22.6 ±0.8	23.2 ±0.8	132	−52	5
103	Ittiri	SI_1	pl	0.920	2.606	70.6	19.1 ±0.5	19.6 ±0.5	333	43	2
			bt	7.850	24.05	80.0	20.6 ±0.5	21.2 +0.5			
26	Padria	SA_2	pl	0.350	0.9125	35.3	17.6 ±0.7	18.0 ±0.8	174	−41	3
27	Padria	SA_2	pl	0.360	0.9758	40.9	18.3 ±0.7	18.7 ±0.7	172 182	−32 −47	17 cvc
30	Bosa	SA_2	pl	0.295	0.8266	30.4	18.9 ±0.7	19.4 ±0.7	139	−40	6
134	Romana	t	pl	2.860	7.288	75.7	17.2 ±0.4	17.6 ±0.5	342	55	1
			bt	8.08	23.22	83.0	19.3 ±0.5	19.9 ±0.5			
133	Banari	SI_2	pl	1.170	2.982	63.0	17.2 ±0.5	17.6 ±0.5	328	48	1
			bt	7.860	21.77	46.0	18.6 ±0.5	19.1 ±0.5			
148	Banari	SI_2	pl	0.612	1.743	53.4	19.2 ±0.5	19.7 ±0.5	341	53	2
97	Banari	SI_2	pl	1.170	3.322	75.5	19.1 ±0.5	19.6 ±0.5	359	52	3
69	Perdaxius	α	pl	0.253	0.6974	30.0	18.6 ±0.7	19.0 ±0.7	179	−47	4
132	Romana	SA_3,α_3	pl	0.323	0.7974	21.6	16.6 ±0.8	17.1 ±0.8	160	−56	3
			pl (HF)	0.365	0.9571	4.6	17.7 ±1.0	18.1 ±1.1			
16	S. Salvatore	β	pl	0.415	0.1867	20.2	3.04±0.18	3.12±0.19	198	−48	5
55	Alghero	porfiriti	pl	0.305	2.164	64.7	47.4 ±1.6	48.6 ±1.7	139 140	−40 −60	11 cvc
106	Valverde	porfiriti	pl	0.216	1.013	20.0	31.5 ±1.5	32.3 ±1.5			
			pl	0.285	1.462	34.4	34.4 ±1.4	35.3 ±1.4	116 147	−21 −38	9 cvc
71	Siliqua	α	pl	0.330	2.232	45.0	45.2 ±1.6	46.4 ±1.6	75	−55	4
			hb	0.615	2.869	63.0	31.3 ±0.9	32.1 ±0.9			
75	Perdaxius	α	pl	0.387	1.475	46.4	25.6 ±0.9	26.3 ±0.9	130	−59	5
			hb	0.433	1.468	50.8	22.8 ±0.8	23.4 ±0.8			
			hb	0.433	1.496	36.3	23.2 ±0.9	23.8 ±0.9			

98	Banari	SA_1	pl (NM)	0.025	0.0928	5.36	25.0 ±4.0	25.6 ±4.1	145	−43	4
			pl (M)	0.100	0.3175	10.45	21.4 ±1.6	21.9 ±1.7	147	−57	cvc
131	Banari	SA_1	pl	0.077	0.2782	15.4	24.3 ±2.0	24.9 ±2.0	160	−35	–
46	Banari	SA_1	–	–	–	–	–	–	163	−33	4
22	Montresta	SA_1	pl	0.243	0.5903	9.5	16.4 ±0.8	16.8 ±0.9	160	−43	2
23	Montresta	SA_1	WR	1.300	3.406	72.2	17.6 ±0.5	18.1 ±0.5	145	−59	5
			pl	0.353	0.9736	27.4	18.6 ±0.9	19.1 ±0.9			
24	Montresta	SA_1	pl (NM)	0.067	0.2053	5.9	20.6 ±1.6	21.2 ±1.6	142	−45	6
			pl (M)	0.230	0.6233	32.3	18.2 ±0.7	18.7 ±0.7			
116′	Osilo	α	pl (NM)	0.110	0.7617	14.1	46.3 ±2.0	47.5 ±2.0	343	53	3
			pl (M)	0.880	2.646	67.5	20.2 ±0.7	20.8 ±0.7			
147	Osilo	α	pl	0.330	1.494	55.2	30.4 ±1.1	31.2 ±1.1	9	44	6
5	Castelsardo	α	pl	0.640	2.003	60.2	21.1 ±0.7	21.6 ±0.8	356	−17	5
11	Castelsardo	α	pl	0.360	1.225	18.5	22.9 ±0.8	23.5 ±0.9	330	50	9
12	Castelsardo	α	–	–	–	–	–	–	332	49	6
1	Castelsardo	$M_1\tau$	pl	0.610	1.870	50.0	20.6 ±0.7	21.2 ±0.7	286 314	53 58	6 6 (2)
2	Castelsardo	$M_1\tau$	pl	0.590	1.782	57.0	20.3 ±0.7	20.9 ±0.7	287	54	2
8	Castelsardo	$M_1\tau$	pl	0.580	1.827	21.4	21.3 ±0.9	21.7 ±0.9	281 305	60 65	3 (2)
72 [a]	Villamassargia	α	WR	1.14	3.213	44.6	18.8 ±0.8	19.3 ±0.8	333	31	4
			WR	1.32	2.677	32.8	13.5 ±0.6	13.9 ±0.6			
			WR	1.59	4.507	32.8	19.1 ±0.8	19.6 ±0.8			
			WR	1.01	3.257	42.5	21.5 ±0.9	22.0 ±0.9			
38 [b]	Sorso	α	WR	2.82		79.6	17.2 ±0.8	17.7 ±0.8	358	48	4
39 [b]	Sorso	α	WR	2.73		61.1	18.9 ±1.0	19.4 ±1.0	157	−36	4
41 [b]	Chiaramonti	α	WR	1.77	unpublished	28.3	18.8 ±1.0	19.3 ±1.0	118 155	−21 −53	17 *cvc*
29 [b]	C. Marargiu	SA_1, α_1	WR	0.78		50.2	23.8 ±1.2	24.4 ±1.2	141	−68	7
118 [c]	Costa Paradiso	τ	WR	3.66	10.75	66.63	16.6 ±0.5	17.0 ±0.5	144	−44	2
119 [c]	Costa Paradiso	τ	WR	4.11	12.51	60.23	17.2 ±0.5	17.7 ±0.5	146	−44	4

Rock type: α=undifferentiated andesites; α_1, SA_1=lower andesites; τ_1, SI_1=lower ignimbrites; α_2, SA_2=upper andesites; $M_1\tau$=ignimbrites interbedded with Aquitanian or Langhian tuffites (M_1t); τ_2, τ_3, SI_2=upper ignimbrites; t=tufs; α_3, SA_3=terminal andesites; β=Plio-Pleistocene basalts. Minerals: pl=plagioclase; hb=hornblende; bt=biotite; WR=whole rock. (HF: treated with hydrofluoric acid; M: magnetic fraction (relative); NM: non-magnetic fraction). Age: (a) $\lambda_\epsilon = 0.585\times10^{-10}$ a^{-1}, $\lambda_\beta = 4.72\times10^{-10}$ a^{-1}, $^{40}K/K_{total} = 1.19\times10^{-4}$ mol/mol [15]; (b) $\lambda_\epsilon = 0.581\times10^{-10}$ a^{-1}, $\lambda_\beta = 4.962\times10^{-10}$ a^{-1}, $^{40}K/K_{total} = 1.167\times10^{-4}$ mol/mol [23]. D=declination; I=inclination; α_{95}=Fisher's statistic parameter; cvc: direction from converging circles; (2): secondary magnetization.
The paleomagnetic data of samples 146, 147, 143 and 145 are original. The others are from Edel [9]. [a] K-Ar data from Savelli et al. [29]. [b] K-ar data from Bellon [27]. [c] K-Ar data from Brotzu et al. [28].

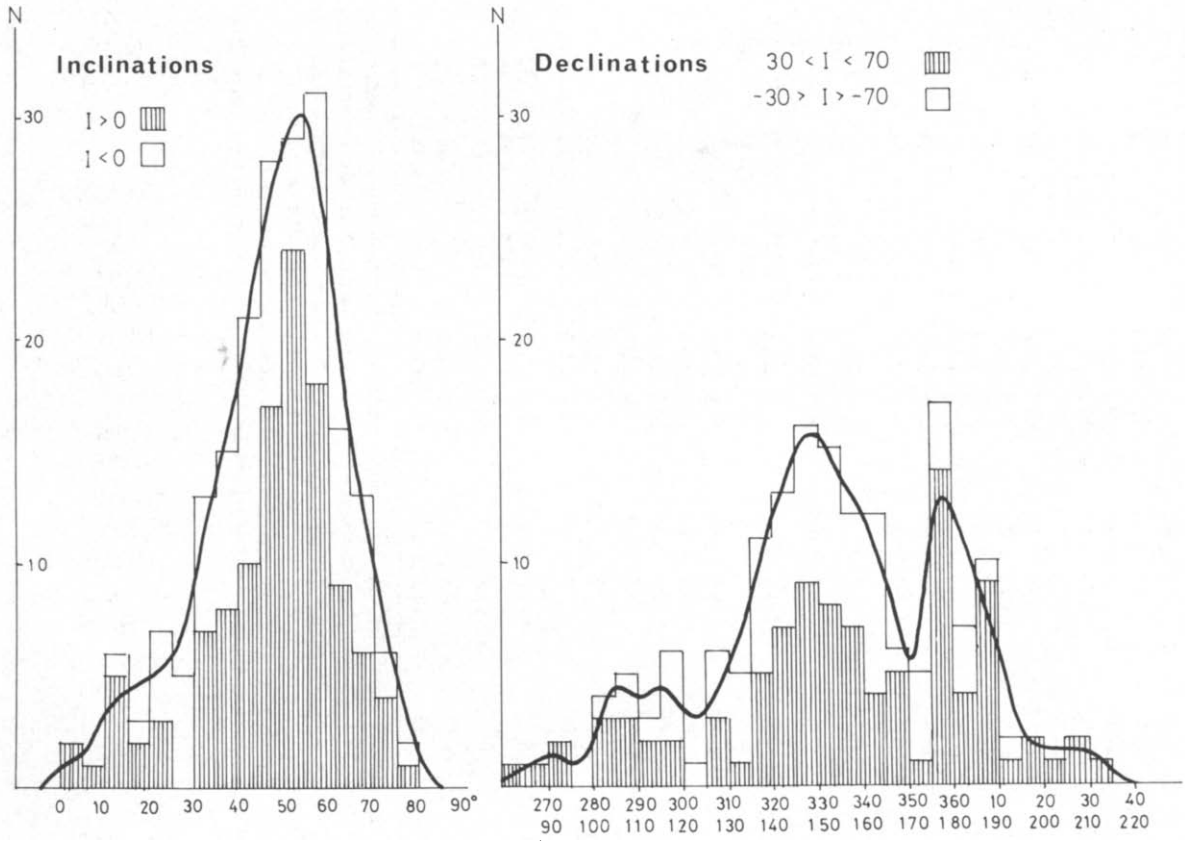


Fig. 3. Histogram of inclinations (left) and declinations (right) including selected published [1–4,6,11–13] and own results (N samples >3; $\alpha_{95} < 20$) from the calc-alkaline volcanics.

geomagnetic virtual pole position is defined as $\phi = 270°E$, $\lambda = 64°N$. The third group ($N = 40$) displays a cluster with a mean direction of $D = 0°$, $I = 50°$, $\alpha_{95} = 3°$. The corresponding pole position is then $\phi = 189°E$, $\lambda = 81°N$. The last two directions, defined by a large number of data, are considered significant. Radiometric dates and a close examination of the stratigraphical record, however, are required to interpret them geologically.

3. K-Ar dating

The K-Ar investigation was carried out on separated minerals, plagioclases and few biotites and hornblendes. We chose not to analyse the whole rocks for several reasons. The whole rock method requires a severe selection of samples by petrography, as exemplified by previous geochronological studies on Tertiary volcanics from Sardinia [8,26–29]. Our sampling, on the other hand, was entirely dependent on paleomagnetic data. Since most of the rocks we analysed, showed evidences of moderate to strong water-rock interaction, we were compelled to extract fresh mineral separates. The ignimbritic rocks are suspected to contain xenoliths of various nature which are deeply altered. The geological significance of K-Ar ages is easier to evaluate on a single system as plagioclase than on a whole rock which represents a complex assemblage of systems [15].

As most of the minerals analysed in this study are plagioclases, their main characteristics as K-Ar chronometers should, therefore, be briefly mentioned. Plagioclases from volcanic rocks are usu-

ally regarded as reliable K-Ar clocks [15–17]. Two reservations must, however, be made to this statement: (1) they may contain excess argon [17,18]; (2) they are sensitive to tectonics [19] which can significantly alter their K-ar ages. The effect of tectonics can be precluded in the age of Sardinia, since the island has not undergone intense folding or faulting after the eruption of Tertiary volcanics.

3.1. Analytical method

The 100–160 μm sieve fraction of the ground rock samples was selected for magnetic separation of feldspars, micas and amphiboles. Whenever necessary, further refinement was done with heavy liquids. Our analytical procedures have been described in detail elsewhere [20]. Briefly, potassium was measured by flame photometry with lithium internal standard [21]. Argon was extracted in a bakeable glass vacuum apparatus and determined by isotope dilution techniques (using ^{38}Ar as a tracer) in a MS 10S mass spectrometer. All samples were treated by the static method. The results are listed in Table 1. Plus-minus figures were estimates of analytical precision at one standard deviation. They were calculated following the procedure given by Cox and Dalrymple [22]. Dates were calculated with both sets of decay constants [15,23]. Since in the last part of the paper, reference is made to the geochronological time scale, the ages used throughout the discussion are those calculated with the former set of constants [15]. The introduction of new constants [23] does not allow a simple translation of the chronological limits of geological stages, unless it is supported by thorough stratigraphical and paleontological arguments.

3.2. Results

Hornblendes. Two hornblendes from the southwest, Cixerri and Sulcis, have been analysed along with their coexisting feldspars. In both cases (sites 71, 75), the hornblendes yield age values (31.3 and 23.2 Ma) lower than those of the feldspars (45.2 and 25.6 Ma). When taking into account the respective Ar retentivenesses of the two systems, the age discrepancies can only be explained by occurrences of excess argon in the plagioclases. It is likely that the hornblende ages reflect times of crystallization although no definite control by stratigraphy is available to substantiate this assumption.

Biotite. Three biotites (sites, 103, 133, 134) from ignimbritic material give K-Ar ages (20.6, 18.6 and 19.3 Ma) higher than those of the coeval plagioclases (19.1, 17.2 and 17.2 Ma). These slight to moderate age differences remain unexplained, since coexisting "high-temperature" feldspars and biotites from Tertiary volcanics yield K-Ar ages in perfect agreement [16,24]. The biotite dates, however, have to be regarded as representative of the crystallization ages, because this mineral is one of the most reliable K-Ar chronometers [15].

Plagioclases. In addition to the previously mentioned plagioclase (sites 71 and 75) showing disagreement of ages with coexisting hornblende, two other feldspars (sites 3,55) show evidences of excess argon. Thus, the stratigraphical record unambiguously allows to eliminate the age (69.8 Ma) yielded by sample from site 3 as being anomalously too high. In case of porfiriti 55 which gives an higher date (47.4 Ma) than porfiriti 106 (34.4 Ma), the occurrence of excess argon is not distinctly demonstrated. K-Ar investigations, however, on rocks from Esterel gave ages between 32 and 35 Ma [25]. These rocks display strong petrological similarities with those of porfiriti. The contemporaneity between the two rocks is substantiated by our result on porfiriti 106. Consequently, one can assume that the plagioclase of sample 55 contains excess argon. On the other hand, the plagioclase of site 22 gives an age that is probably too low, compared to other contemporaneous samples from the same area.

The remaining feldspars are assumed to give ages that are geologically meaningful. Two arguments are proposed to support this contention: the results are in conformity with the relative stratigraphy [6,26]; the plagioclases analysed in this study display low K_2O content and may, therefore, be expected to retain argon completely [15,16,24].

When taking into account the analytical data that are geologically significant, the following re-

marks can be made from our K-Ar investigation:

(1) A good agreement exists between the absolute dates and the relative stratigraphy [6,26] of the ignimbrites. The two parts of the series are chronologically well defined; the lower ignimbrites (SI_1, $M_1\tau$) display ages ranging from 20.3 to 22.6 Ma while the upper ones (SI_2, τ) give dates comprised between 18.3 and 19.2 Ma. As each part yields paleomagnetic directions that are definitely different from one another, the ignimbritic series appears to provide severe constraints on the determination of the time of rotation.

(2) The Tertiary volcanism in Sardinia lasted from 34 to 16.5 Ma. Previous K-Ar investigations [8,26–29] on those volcanics concluded to a 30–13.5 Ma interval for the event. These slightly different conclusions can be easily ascribed to analyses of samples covering different stratigraphic intervals. For instance, we have focused our attenuation on ignimbritic material and porfiriti, and not included the uppermost andesites (SA_3), whereas other workers [8,26–29] have well documented the andesitic formations. There is, therefore, no fundamental contradiction among the different sets of data, rather they complement one another.

4. Discussion: correlation K-Ar dating, paleomagnetism and stratigraphy

All the available K-Ar data, that are assumed to correspond to our paleomagnetic sites (Table 1) are taken into account in the correlation. In Fig. 4 the declinations are plotted against the corresponding K-Ar dates. Most of the sites fall within a 30 to 40° wide zone, which is equivalent to the magnitude of paleosecular variations of the Earth magnetic field. Two anomalous field directions, however, can be observed. The first one is displayed by andesitic sites of Cixerri (site 71), Sulcis and Sarroch [9,10,30] which show declinations around 270°N for the normal field directions and 90°N for the reversed [4,9,10,30]. These sites probably correspond to the lowest terms of the andesitic volcanism. Their specific directions suggested that southwest Sardinia might have undergone motions independently from the rest of the Corso-Sardinian microplate [10,30]. More data are needed to support this hypothesis. The second anomalous direction is one of the $M_1\tau$ ignimbritic layer interbedded with $M_1\tau$ tuffites of Castelsardo (sites 1, 2, 8). The good quality of paleomagnetic data corresponding to these sites suggests the occurrence of anomalous field directions during the cooling of the lavas. In the time-span of 35 to 20.5 Ma the lower andesites of Bosano, Logudoro, Anglona and some of Cixerri and Sulcis, display declinations around a mean direction of 329°N. Between 20.5 and 19 Ma, the declinations of the upper andesites and Logudoro, Bosano and Anglona, of the upper ignimbrites of Logudoro and of the tuffites (M_1t) of Anglona exhibit northwest to north directions. The upper andesites of Bosano, the uppermost layers of the upper ignimbrites of Logudoro and the upper ignimbrites of Castelsardo display north declinations. An identical direction is shown by an andesitic dike (site 59) that intrudes the lower andesites of Sulcis. This direction is also given by the Plio-Pleistocene alkali basalts [31–33] dated at 2.3 to 2.8 Ma [26,27]. The andesite of San Salvatore (site 16) yielding an age of 3 Ma should, therefore, belong to the same volcanic cycle.

The analytical uncertainties of age determinations and the shortness of the reversals do not usually allow a correlation between the polarity of magnetizations and the geomagnetic reversal time scale of the Oligocene-Miocene. Nevertheless, the attempt may be worthwhile in two particular cases. The ignimbritic layer $M_1\tau$ interbedded with tuffites, has been sampled at various localities. It is featured by an unusual WNW declination and a positive inclination, while its age is well determined between 20.3 and 21.2 Ma. The normal polarity can be correlated with the relative long period that corresponds to anomaly 6 [34–36]. The drift of the declinations towards the north has taken place right after the emplacement of the $M_1\tau$ ignimbrites [10]. The correlation is tenuous between the positive inclination of the upper ignimbrites from Castelsardo and Logudoro (SI_2, τ_2), and the anomaly 5d of the time scale, which would correspond to the drift-end.

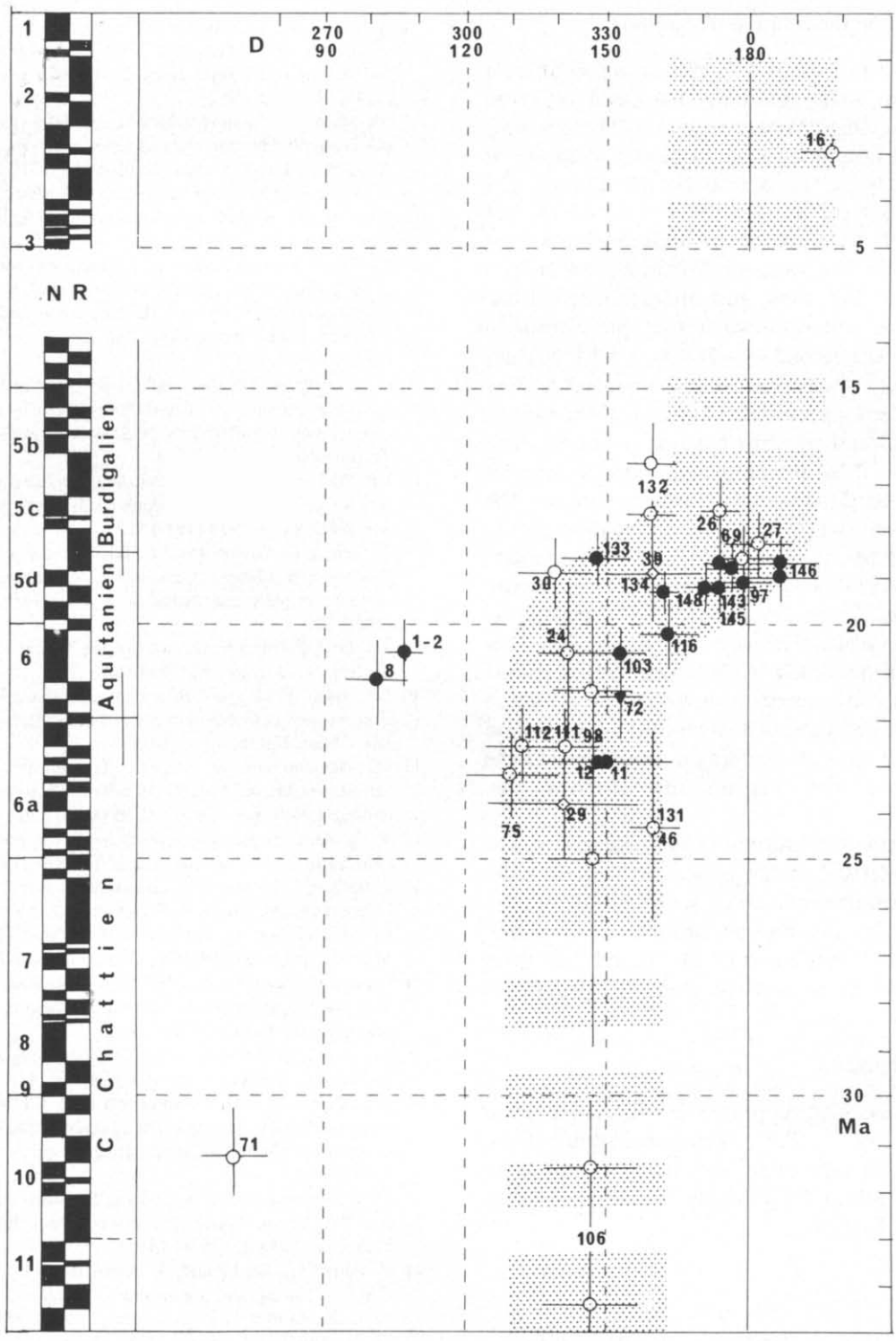


Fig. 4. Oligo-Miocene drift of Sardinian declinations after elimination of doubtful dating from Table 1. Declination ($\alpha_{95}/\cos I$) versus age ($\pm 1\sigma$).● = normal polarity; ○ = reversed polarity. Magnetic time scale taken from Heirtzler et al. [34], McKenzie and Sclater [35], Schlich [36] and stratigraphic time scale from Odin [37].

5. Conclusion: the rotation of Sardinia

In stable Europe and in Africa, no significant drift of magnetic directions has been observed during the Oligo-Miocene period. The geomagnetic poles computed from published data are as follows: $\phi = 150°E$, $\lambda = 80°N$ for Europe (23 studies); $\phi = 162°E$, $\lambda = 83°N$ for Africa (11 studies) [10]. Two poles have been determined for Sardinia. The first one, $\phi = 270°E$, $\lambda = 64°N$, corresponds to the lower and middle series of the calc-alkaline volcanism which took place from 34 to 20 Ma. The second, $\phi = 189°E$, $\lambda = 81°N$ characterizes the uppermost series, emplaced around 19 Ma. The long period, 14 Ma, with northwest declinations and the drift towards the north must be interpreted by an counterclockwise rotation of 30° of the Sardinian block relative to Europe. The phenomenon occurred within a rather short period of time, 1.5 Ma at most, in the Upper Aquitanian, i.e. 20.5 to 19 Ma [37]. The event may be placed between the 6 and 5d anomalies of the geomagnetic time scale. The age of the rotation determined in this study is, therefore, older than the one obtained by a previous investigation [8] which has ascertained the phenomenon between 17 and 15 Ma. The ways of sampling as underlined in the introduction, may explain the observed discrepancy.

The mean inclination $I = 50°$, $\alpha_{95} = 3°$ ($N = 40$), characteristic of the upper parts of the calc-alkalic volcanism indicates a paleolatitude of 31°. Consequently, the present latitude of Sardinia, which is 40°, implies that the island has been moving towards the north by about 1000 km in the last 18 Ma.

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